Endalk Y. Gebru

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EDUCATION

University of Minnesota – Twin Cities *Minneapolis, MN*
Electrical and Computer Engineering, PhD *Expected: 2028*

- CGPA: 3.97 | Advisor: Dr. Sachin S. Sapatnekar
- Relevant Coursework: RF IC Design, Analog IC Design, VLSI Design, VLSI Design Automation, Heterogeneous Integrated Circuits, Design Automation Techniques for Variation-Aware Computing, Machine Learning, Deep Learning, Computational Matrix Theory

Addis Ababa University *Addis Ababa, Ethiopia*
Electrical and Computer Engineering, BSc *May 2022*

- GPA: 4.0 | Thesis: Long-Range Visible Light Communication System Design and Implementation
- Selected Coursework: RF/Microwave Propagation, Analog /Digital Electronics, Computer Architecture, Data Structures/Algorithms

WORK EXPERIENCE

Analog EDA R&D Intern | Cadence Design Systems *(San Jose, CA, USA)* *May 2025 – Aug 2025*

- Developed a local LLM-based workflow for circuit design in Cadence Virtuoso, combining retrieval-augmented generation (RAG) across multiple documents with YOLO-based circuit detection in papers and books
- Curated a circuit database and used retrieved circuit information to generate netlists for instantiation in the design flow

Graduate Research Assistant | University of Minnesota *(Minnesota, USA)* *Aug 2023 – present*

- Develop analog circuit automation and physical design frameworks using classical algorithms and optimization methods
- Apply AI/ML techniques to circuit synthesis, layout optimization, and automated design workflows for analog and mixed-signal circuits

Research Assistant | McGill University *(Montreal, Canada | Remote, full-time)* *Jun 2022 – Aug 2023*

- Developed a MATLAB-based wind engineering and uncertainty quantification framework
- Worked on PC-NARX modeling of stochastic wind response and a tuned mass damper (TMD) simulator

TECHNICAL SKILLS

AI/ML: PyTorch, scikit-learn, LLMs (Llama, Qwen, GPT, Gemini), RAG, YOLO
EDA/Simulation: Cadence Virtuoso, Keysight ADS, Synopsys VCS, HSPICE, Magic, LTspice, Ngspice, Simulink
Programming & Scripting: Python, C++, Tcl, Git/GitHub, Make, shell scripting, MATLAB, Java, Cadence SKILL

SELECTED PUBLICATIONS & MANUSCRIPTS

[1] **E. Y. Gebru**, R. Harjani, and S. S. Sapatnekar, "Co-MiX: A Co-ordinated Multi-Agent Framework for MiXed-Signal IC Design from Intent to GDSII," *Design, Automation & Test in Europe Conference (DATE)*, 2027 (submitted).

[2] **E. Y. Gebru**, Ramprasath S., R. Harjani, and S. S. Sapatnekar, "Row-Based Layout Synthesis for Analog Circuits Using Height-Quantized Primitives," *IEEE Trans. Computer-Aided Design of Integrated Circuits and Systems (TCAD)* (2nd-round review).

[3] S. Ghosh, S. S. Peri, Ramprasath S., S. A. Berhan, **E. Y. Gebru**, R. Harjani, and S. S. Sapatnekar, "COmPOSER: Circuit Optimization of mm-wave/RF circuits with Performance-Oriented Synthesis for Efficient Realizations," in *Proc. ACM/IEEE Design Automation Conference (DAC)*, 2026.

[4] S. A. Berhan, **E. Gebru**, S. Ghosh, J. Poojary, S. S. Peri, J. Hu, R. Harjani, Ramprasath S., and S. S. Sapatnekar, "Automated Analog Layout Synthesis," *J. Indian Inst. Sci.*, 2026.

[5] S. A. Berhan, **E. Y. Gebru**, S. Ghosh, S. S. Peri, R. Harjani, and S. S. Sapatnekar, "Special Research Session: The Quest for the Optimal Use of AI in AMS/RF Design," in *Proc. ACM/IEEE Design Automation Conference (DAC)*, 2026.

[6] **E. Y. Gebru**, S. Ghosh, R. Harjani, and S. S. Sapatnekar, "Parasitics-Aware Framework for Integrated OTA Sizing and Layout Synthesis in FinFET Technologies," in *Proc. Int. Symp. Quality Electronic Design (ISQED)*, 2026.

[7] S. Ghosh, **E. Y. Gebru**, C. V. Kashyap, R. Harjani, and S. S. Sapatnekar, "Accelerating OTA Circuit Design: Transistor Sizing Based on a Transformer Model and Precomputed Lookup Tables," in *Proc. Design, Automation & Test in Europe (DATE)*, 2025.

PROJECTS

- [EDA] **CMOS Readout IC (ROIC) Layout Automation**
  Automated infrared bolometer ROIC layouts using SkyWater 130 nm and ALIGN in collaboration with Nanohmics Inc.
- [EDA] **Static Timing Analysis (STA)**
  Built a C++ STA framework from scratch using NLDM timing models to compute circuit delays, slack, and critical paths
- [EDA] **FastPlace Placement Framework**
  Built a placement framework from scratch implementing the FastPlace algorithm; tested it on IBM benchmark circuits
- [ML & Optimization] **Sensitivity-Guided Hierarchical Optimization for Chiplet-Based GPT-2 Mapping**
  Developed a parametric sensitivity-guided hierarchical optimization framework using HISIM 2.0 to minimize the cost–latency product for GPT-2 mapping under an area budget
- [ML & Optimization] **Diffusion Models for Out-of-Distribution Inference**
  Investigated diffusion models for inference on inputs outside the training distribution
- [Design] **94 GHz PA and LNA Design in Keysight ADS**
  Designed and simulated PA/LNA circuits targeting 4 GHz bandwidth, ≥10 mW radiated power, and receiver NF ≤5 dB, balancing PA efficiency and LNA power
- [Design] **Fully Differential CMOS Biquad Filter**
  Designed and implemented a low-power biquad filter for hearing aids using 50 nm technology
- [Design] **14-bit General-Purpose Microprocessor**
  Designed a microprocessor with microprogrammed control and an 18-instruction set; implemented in Proteus

ACHIEVEMENTS

- **2022 Addis Ababa University Gold Medal Award:** Awarded for academic excellence to the program's top-ranking undergraduate
- **POSCO African Fellowship Award (2020–2022):** three-year financial award for outstanding academic performance